COMPUTER ENGINEERING

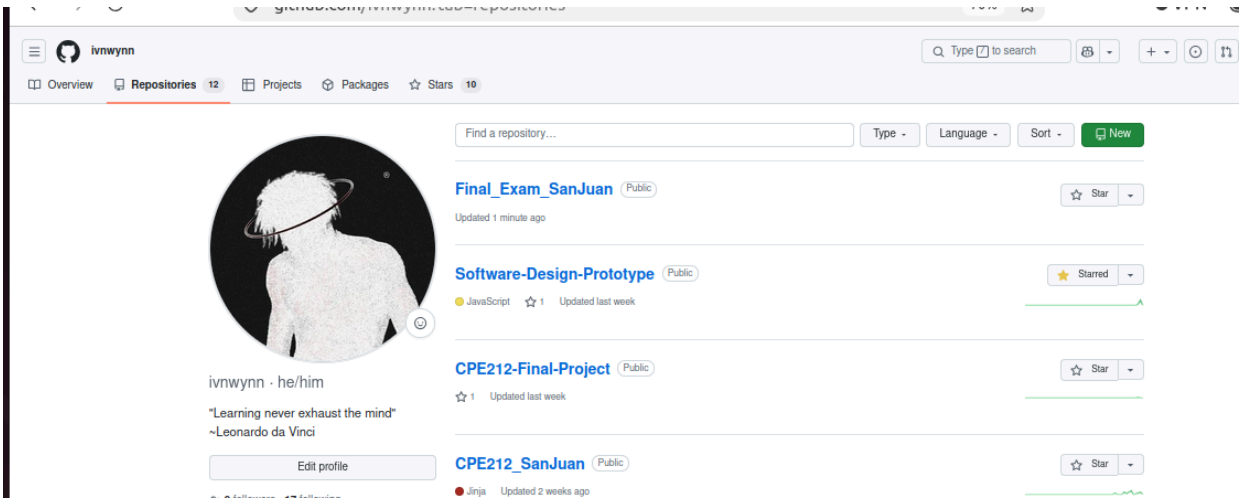
QUEZON CITY

COURSE CODE/TITLE:	CPE212	**SECTION:**	CPE31S2
DATE PERFORMED:	10/29/26	**DATE SUBMITTED:**	10/29/26
NAME:	Ivan San Juan	**INSTRUCTOR:**	Engr. Jimlord Quejado

ACTIVITY NO. **Hands-on Skill Exam**

1. PROCEDURE OUTPUT

1. Create a repository and label it as "Final_Exam_Surname"



2. Clone your new repository

```
sanjuan@LocalMachine:~$ gh repo create Final_Exam_SanJuan --public --clone
✓ Created repository ivnwynn/Final_Exam_SanJuan on GitHub
hint: Using 'master' as the name for the initial branch. This default branch nam
e
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/sanjuan/Final_Exam_SanJuan/.git/
sanjuan@LocalMachine:~$ ls
 ansible-lab      Downloads             Pictures    Software-Design-Prototype
 CPE212_SanJuan   Final_Exam_SanJuan    Portfolio   Templates
 Desktop          Midterm-Exam-CPE212   Public     'Untitled Document 1'
 Documents        Music                 snap        Videos
sanjuan@LocalMachine:~$ cd Final_Exam_SanJuan
sanjuan@LocalMachine:~/Final_Exam_SanJuan$ ls -a
.  ..  .git
sanjuan@LocalMachine:~/Final_Exam_SanJuan$
```

in your VM

3. Create an Ansible playbook that does the following with an input of a config.yaml file and structure inventory file.

1. 3.1 Install and configure one enterprise service that can be installed in Debian and Centos servers

```yaml
- name: Install and configure enterprise service
  hosts: debian_servers:centos_servers
  become: true
  vars_files:
    - config.yaml
  tasks:
    - name: Install enterprise service package (Debian)
      ansible.builtin.apt:
        name: "{{ enterprise_service.package_debian }}"
        state: present
        update_cache: true
      when: ansible_os_family == "Debian"

    - name: Install enterprise service package (RedHat)
      ansible.builtin.dnf:
        name: "{{ enterprise_service.package_redhat }}"
        state: present
      when: ansible_os_family == "RedHat"

    - name: Ensure enterprise service is enabled and running
      ansible.builtin.service:
        name: "{{ enterprise_service.service_name }}"
        state: started
        enabled: true
```

config.yml

```yaml
# config.yaml
enterprise_service:
  name: nginx
  package_debian: nginx
  package_redhat: nginx
  service_name: nginx

```

```
sanjuan@LocalMachine:~/Final_Exam_SanJuan$ ansible-playbook -i inventory/host.yml site.yml -K
BECOME password:

PLAY [Load configuration variables] ********************************************************************************

TASK [Ensure config is loaded (no-op)] *****************************************************************************
ok: [localhost] => {
    "msg": "Configuration loaded successfully"
}

PLAY [Install and configure enterprise service] ********************************************************************

TASK [Gathering Facts] *********************************************************************************************
[WARNING]: Platform linux on host centos-node is using the discovered Python interpreter at /usr/bin/python3.9, but future
installation of another Python interpreter could change the meaning of that path. See https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
ok: [centos-node]
[WARNING]: Platform linux on host ubuntu-node is using the discovered Python interpreter at /usr/bin/python3.10, but future
installation of another Python interpreter could change the meaning of that path. See https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
ok: [ubuntu-node]

TASK [Install enterprise service package (Debian)] *****************************************************************
skipping: [centos-node]
changed: [ubuntu-node]

TASK [Install enterprise service package (RedHat)] *****************************************************************
skipping: [ubuntu-node]
changed: [centos-node]

TASK [Ensure enterprise service is enabled and running] ************************************************************
changed: [centos-node]
ok: [ubuntu-node]
[WARNING]: Could not match supplied host pattern, ignoring: monitoring_servers
```

2. 3.2 Install and configure one monitoring tool that can be installed in Debian and Centos servers (if it is a stack there should be option of different host)

```yaml
- name: Deploy Munin server
  hosts: "{{ monitoring.munin_server_group }}"
  become: true
  vars_files:
    - config.yaml
  roles:
    - role: monitoring
      vars:
        run_munin_server: "{{ monitoring.run_munin_server }}"
        munin_node_allowed_ips: "{{ monitoring.munin_node_allowed_ips }}"

- name: Deploy Munin node on monitored hosts
  hosts: monitored_hosts
  become: true
  vars_files:
    - config.yaml
  roles:
    - role: monitoring
      vars:
        run_munin_server: false
        munin_node_allowed_ips: "{{ monitoring.munin_node_allowed_ips }}"
```

```
  service_name: nginx

monitoring:
  munin_server_group: monitoring_servers
  run_munin_server: true
  munin_node_allowed_ips: "192\\.168\\.1\\..*"

motd:
```

```
[WARNING]: Could not match supplied host pattern, ignoring: monitoring_servers

PLAY [Deploy Munin server] ****************************************************************************************************
skipping: no hosts matched
[WARNING]: Could not match supplied host pattern, ignoring: monitored_hosts

PLAY [Deploy Munin node on monitored hosts] ***********************************************************************************
skipping: no hosts matched
```

3. 4.4 Change Motd as "Ansible Managed by <username>"

```
- name: Configure MOTD
  hosts: all
  become: true
  vars_files:
    - config.yaml
  tasks:
    - name: Deploy static MOTD
      ansible.builtin.copy:
        content: "{{ motd.managed_by }}\n"
        dest: /etc/motd
        owner: root
        group: root
        mode: '0644'
```

```
motd:
  managed_by: "Ansible Managed by San Juan"
```

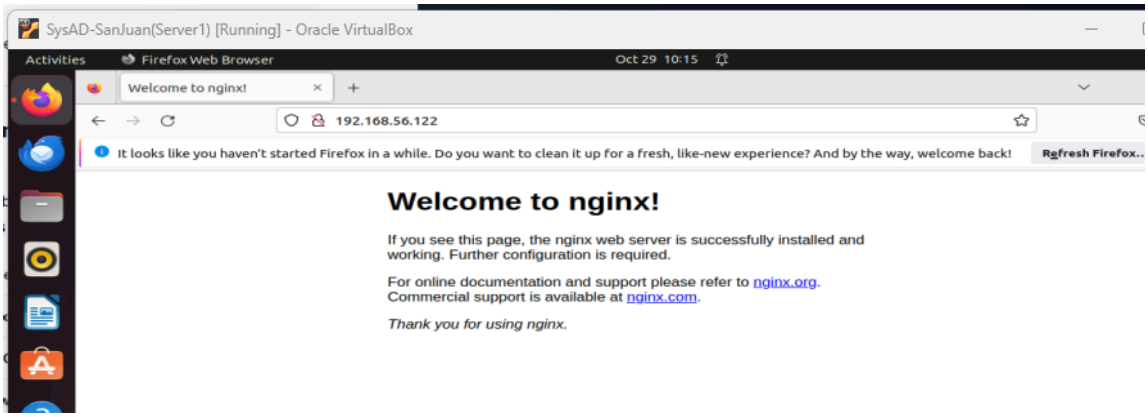
```
PLAY [Configure MOTD] **********************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************
ok: [ubuntu-node]
ok: [centos-node]

TASK [Deploy static MOTD] ******************************************************************************************************************
changed: [ubuntu-node]
changed: [centos-node]

PLAY RECAP *********************************************************************************************************************************
centos-node                : ok=5    changed=3    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
localhost                  : ok=1    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
ubuntu-node                : ok=5    changed=2    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
```

```
sanjuan@LocalMachine:~/Final_Exam_SanJuan$ ansible -i inventory/host.yml all -m shell -a 'systemctl is-active n
in-node' --become
[WARNING]: Platform linux on host ubuntu-node is using the discovered Python interpreter at /usr/bin/python3.10
future installation of another Python interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter_discovery.html for more information
ubuntu-node | CHANGED | rc=0 >>
active
inactive
[WARNING]: Platform linux on host centos-node is using the discovered Python interpreter at /usr/bin/python3.9,
future installation of another Python interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter_discovery.html for more information
centos-node | CHANGED | rc=0 >>
active
inactive
```



4. Push and commit your files in GitHub

```
sanjuan@LocalMachine:~/Final_Exam_SanJuan$ git status
On branch master

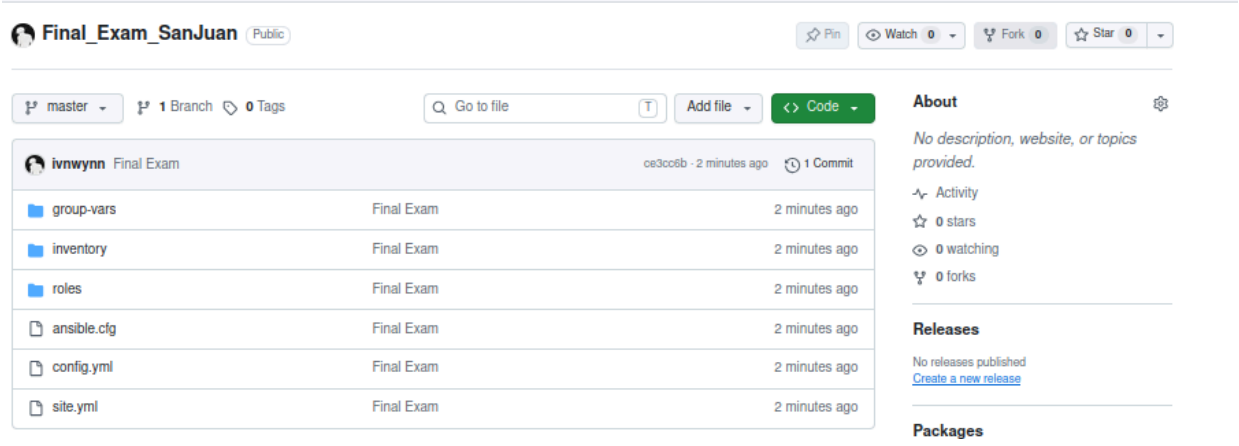
No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        ansible.cfg
        config.yml
        group-vars/
        inventory/
        roles/
        site.yml

nothing added to commit but untracked files present (use "git add" to track)
sanjuan@LocalMachine:~/Final_Exam_SanJuan$ git add .
```

```
sanjuan@LocalMachine:~/Final_Exam_SanJuan$ git commit -m "Final Exam"
[master (root-commit) ce3cc6b] Final Exam
 17 files changed, 255 insertions(+)
 create mode 100644 ansible.cfg
 create mode 100644 config.yml
 create mode 100644 group-vars/all.yml
 create mode 100644 group-vars/debian.yml
 create mode 100644 group-vars/redhat.yml
 create mode 100644 inventory/host.yml
 create mode 100644 roles/enterprise_service
 create mode 100644 roles/monitoring/README.md
 create mode 100644 roles/monitoring/defaults/main.yml
 create mode 100644 roles/monitoring/handlers/main.yml
 create mode 100644 roles/monitoring/meta/main.yml
 create mode 100644 roles/monitoring/tasks/main.yml
 create mode 100644 roles/monitoring/tests/inventory
 create mode 100644 roles/monitoring/tests/test.yml
 create mode 100644 roles/monitoring/vars/main.yml
 create mode 100644 roles/monitorting
 create mode 100644 site.yml
```

```
sanjuan@LocalMachine:~/Final_Exam_SanJuan$ git push --set-upstream origin master
Enumerating objects: 27, done.
Counting objects: 100% (27/27), done.
Delta compression using up to 12 threads
Compressing objects: 100% (15/15), done.
Writing objects: 100% (27/27), 4.10 KiB | 4.10 MiB/s, done.
Total 27 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/ivnwynn/Final_Exam_SanJuan.git
 * [new branch]      master -> master
Branch 'master' set up to track remote branch 'master' from 'origin'.
```



5. For your final exam to be counted, please paste your repository link as an answer in this exam.

https://github.com/ivnwynn/Final_Exam_SanJuan

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or

completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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